Hyrje në algoritme

Algoritmet
dhe
Strukturat e të dhënave

Hyrje në algoritme – pasqyra e lëndës

- Programming and problem solving, with applications.
- Algoritmi: mënyrë për zgjidhjen e një problemi.

- Struktura e të dhënave: mënyrë për të ruajtur informacion.

topic	data structures and algorithms
data types	stack, queue, bag, union-find, priority queue
sorting	quicksort, mergesort, heapsort, radix sorts
searching	BST, red-black BST, hash table
graphs	BFS, DFS, Prim, Kruskal, Dijkstra
strings	KMP, regular expressions, tries, data compression
advanced	B-tree, k-d tree, suffix array, maxflow

Pse studiohen algoritmet?

Algoritmet kanë zbatim të gjerë.

Internet. Web search, packet routing, distributed file sharing, ...

Biologji. Human genome project, protein folding, ...

Komputerë. Circuit layout, file system, compilers, ...

Grafikë kompjuterike. Movies, video games, virtual reality, ...

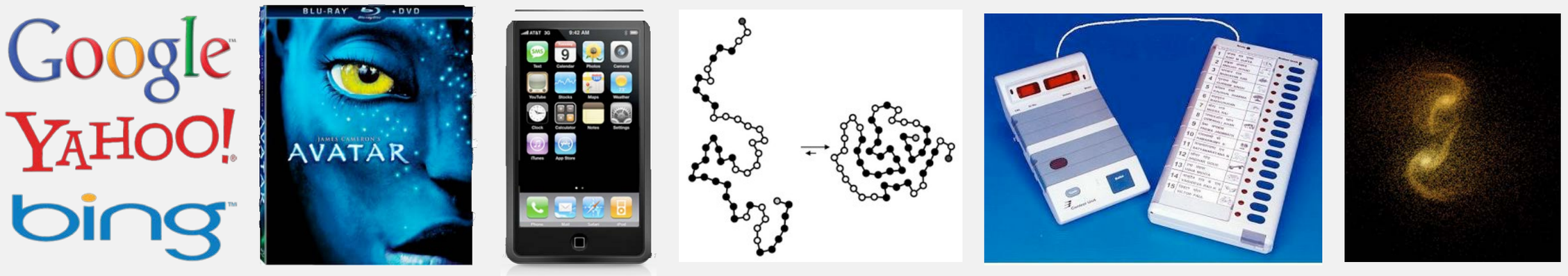
Siguri. Cell phones, e-commerce, voting machines, ...

Multimedia. MP3, JPG, DivX, HDTV, face recognition, ...

Rrjete sociale. Recommendations, news feeds, advertisements, ...

Fizikë. N-body simulation, particle collision simulation, ...

⋮



Pse studiohen algoritmet?

Their impact is broad and far-reaching.

Mysterious algorithm was 4% of trading activity last week

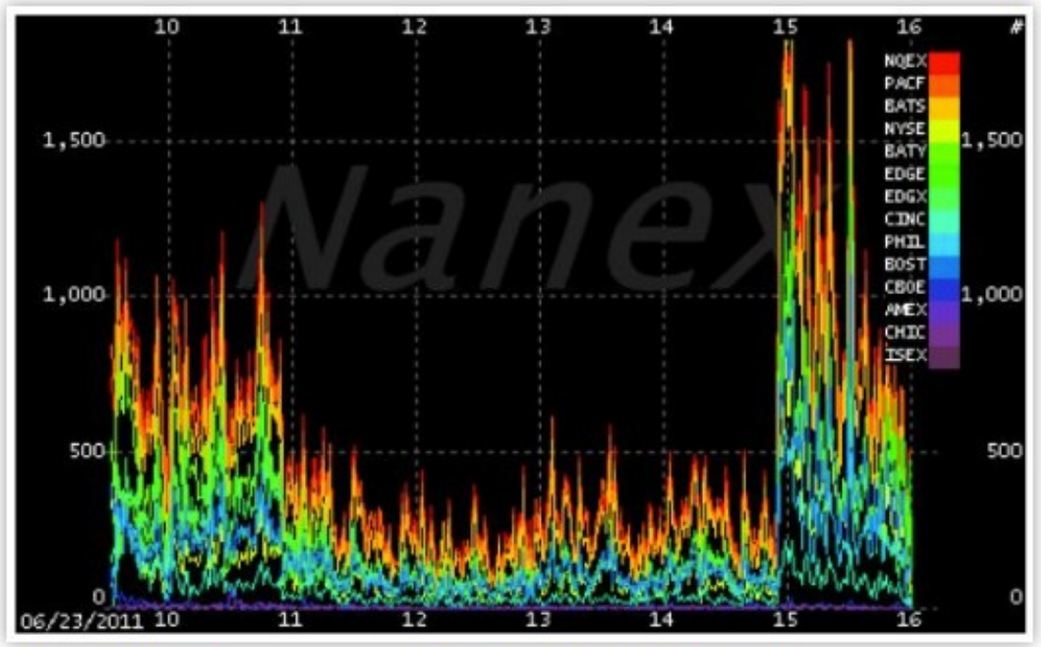
October 11, 2012

A single mysterious computer program that placed orders — and then subsequently canceled them — made up 4 percent of all quote traffic in the U.S. stock market last week, according to the top tracker of high-frequency trading activity.

The motive of the algorithm is still unclear, *CNBC* reports.

The program placed orders in 25-millisecond bursts involving about 500 stocks, according to Nanex, a market data firm. The algorithm never executed a single trade, and it abruptly ended at about 10:30 a.m. ET Friday.

[+]



Generic high frequency rrading chart (credit: Nanex)

"My guess is that the algo was testing the market, as high-frequency frequently does," says Jon Najarian, co-founder of TradeMonster.com. "As soon as they add bandwidth, the HFT crowd sees how quickly they can top out to create latency." (*Read More*: Unclear What Caused Kraft Spike: Nanex Founder.)

Pse studiohen algoritmet?

Për stimulim intelektual.

“ For me, great algorithms are the poetry of computation. Just like verse, they can be terse, allusive, dense, and even mysterious. But once unlocked, they cast a brilliant new light on some aspect of computing. ” — *Francis Sullivan*



FROM THE EDITORS

THE JOY OF ALGORITHMS

Francis Sullivan, Associate Editor-in-Chief

THE THEME OF THIS FIRST-OF-THE-CENTURY ISSUE OF *COMPUTING IN SCIENCE & ENGINEERING* IS ALGORITHMS. IN FACT, WE WERE BOLD ENOUGH—AND PERHAPS FOOLISH ENOUGH—TO CALL THE 10 EXAMPLES WE'VE SELECTED "THE TOP 10 ALGORITHMS OF THE CENTURY."

Computational algorithms are probably as old as civilization. Sumerian cuneiform, one of the most ancient written records, consists partly of algorithm descriptions for reckoning in base 60. And I suppose we could claim that the Druid algorithm for estimating the start of summer is embodied in Stonehenge. (That's really hard hardware!)

Like so many other things that technology affects, algorithms have advanced in startling and unexpected ways in the 20th century—at least it looks that way to us now. The algorithms we chose for this issue have been essential for progress in communications, health care, manufacturing, economics, weather prediction, defense, and fundamental science. Conversely, progress in these areas has stimulated the search for ever-better algorithms. I recall one late-night ball session on the Maryland Shore when someone asked, "Who first ate a pie. After all, they don't look very appetizing." After the usual speculations about the observed behavior of sea gulls, someone gave what must be the right answer—namely, "A very hungry person first ate a crab."

The flip side to "necessity is the mother of invention" is "invention creates its own necessity." Our need for powerful machines always exceeds their availability. Each significant computation brings insights that suggest the next, usually much larger, computation to be done. New algorithms are an attempt to bridge the gap between the demand for cycles and the available supply of them. We've become accustomed to gaining the Moore's Law factor of two every 18 months. In effect, Moore's Law changes the constant in front of the estimate of running time as a function of problem size. Important new algorithms do not come along every 1.5 years, but when they do, they can change the exponent of the complexity!

For me, great algorithms are the poetry of computation. Just like verse, they can be terse, allusive, dense, and even mysterious. But once unlocked, they cast a brilliant new light on some aspect of computing. A colleague recently claimed that he'd done only 15 minutes of productive work in his whole life. He wasn't joking, because he was referring to the 15 minutes during which he'd sketched out a fundamental optimization algorithm. He regarded the previous years of thought and investigation as a sunk cost that might or might not have paid off.

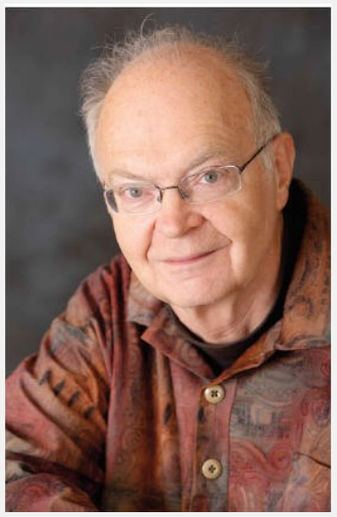
Researchers have cracked many hard problems since 1 January 1900, but we are passing some even harder ones on to the next century. In spite of a lot of good work, the question of how to extract information from extremely large masses of data is still almost untouched. There are still very big challenges coming from more "traditional" tasks, too. For exam- ple, we need efficient methods to tell when the result of a large crab? floating-point calculation is likely to be correct. Think of the way that check sums function. The added computational cost is very small, but the added confidence in the answer is large.

Is there an analog for things such as huge, multidisciplinary optimizations? At an even deeper level is the issue of reasonable methods for solving specific cases of "impossible" problems. Instances of NP-complete problems crop up in attempting to answer many practical questions. Are there efficient ways to attack them?

I suspect that in the 21st century, things will be ripe for another revolution in our understanding of the foundations of computational theory. Questions already arising from quantum computing and problems associated with the generation of random numbers seem to require that we somehow tie together theories of computing, logic, and the nature of the physical world.

The new century is not going to be very restful for us, but it is not going to be dull either!

“ An algorithm must be seen to be believed. ” — *Donald Knuth*



Pse studiohen algoritmet?

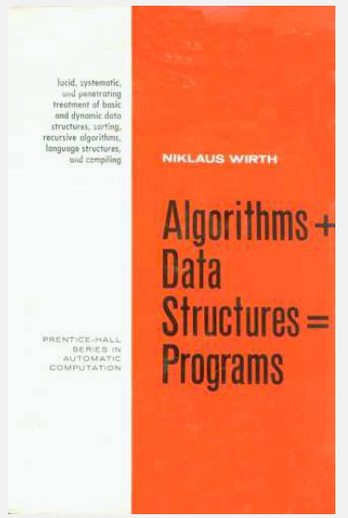
Për t'u bërë programer kompetent.

" I will, in fact, claim that the difference between a bad programmer and a good one is whether he considers his code or his data structures more important. Bad programmers worry about the code. Good programmers worry about data structures and their relationships. "

— Linus Torvalds (creator of Linux)



" Algorithms + Data Structures = Programs. " *— Niklaus Wirth*



Pse studiohen algoritmet?

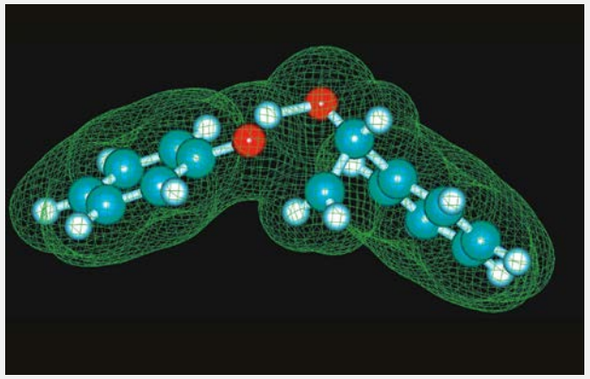
Algoritmet mund të japin përgjigje në fshehtësitë e jetës dhe Universit.

> *“ Computer models mirroring real life have become crucial for most advances made in chemistry today…. Today the computer is just as important a tool for chemists as the test tube. ”*
>
> — *Royal Swedish Academy of Sciences*
>
> (*Nobel Prize in Chemistry 2013*)



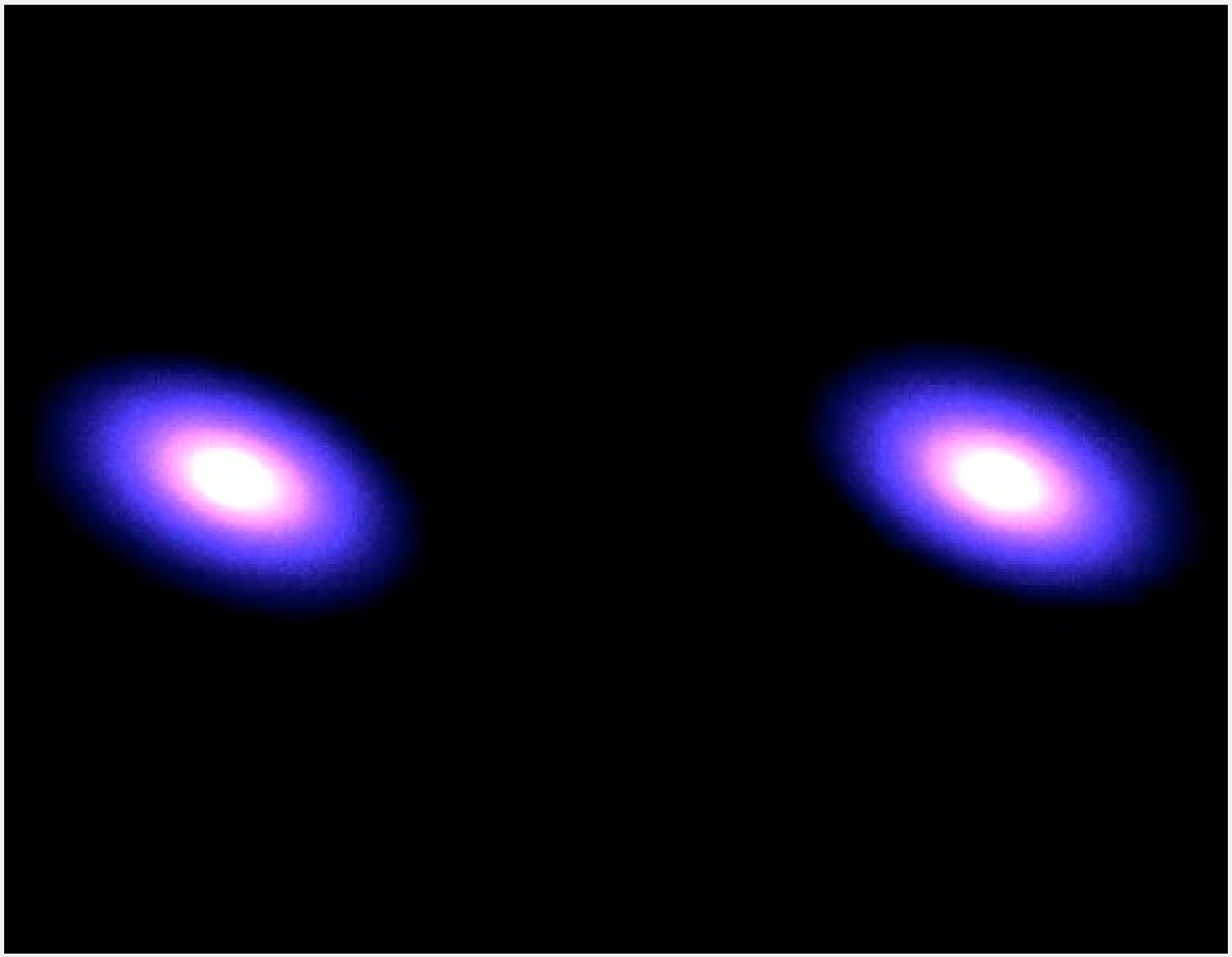


Martin Karplus, Michael Levitt, and Arieh Warshel



Pse studiohen algoritmet?

Për të zgjidhur problemet të cilat përndryshë nuk mund të adresohen.



http://www.youtube.com/watch?v=ua7YlN4eL_w

Pse studiohen algoritmet?

For fun and profit.

Google

Apple Computer

facebook

Cisco Systems

Nintendo

IBM

Jane Street

Morgan Stanley

NETFLIX

Adobe

RSA Security

D E Shaw & Co

ORACLE

Pandora

Akamai

YAHOO!

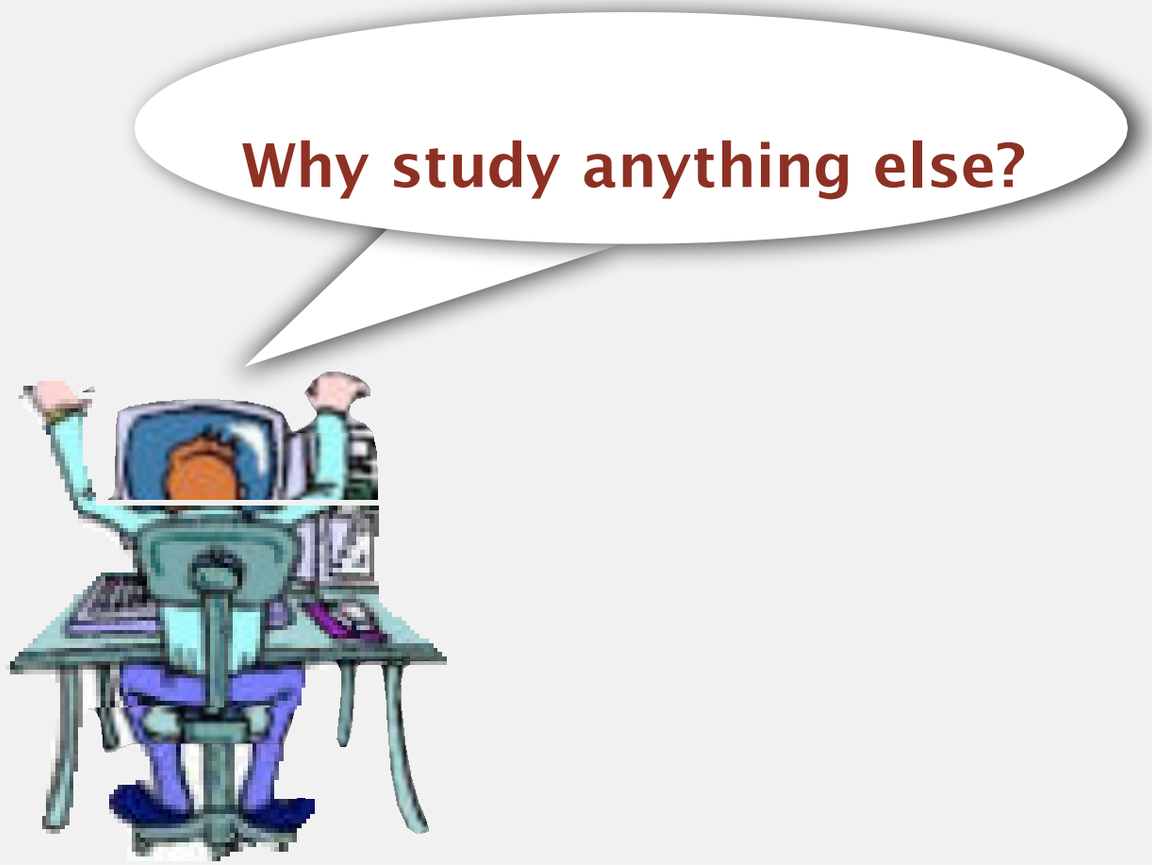
amazon.com

Microsoft

Pixar Animation Studios

Pse studiohen algoritmet?

- Algoritmet kanë zbatim të gjerë
- Old roots, new opportunities.
- Për stimulim intelektual.
- Për t'u bërë programmer kompetent.
- Algoritmet mund të japin përgjigje në fshehtësitë e jetës dhe Universit.
- Për të zgjidhur problemet të cilat përndryshë nuk mund të adresohen.
- For fun and profit.



Ligjëratat

Prezentime. Materialet e librit + diskutime.

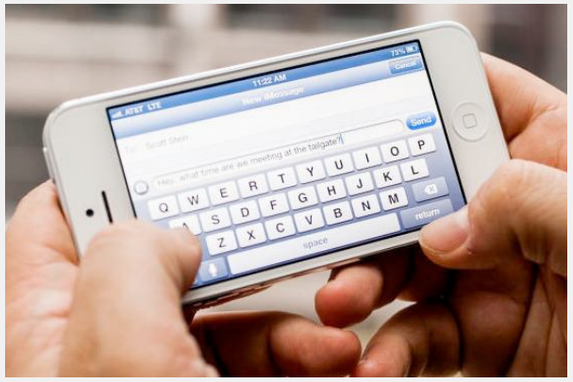
Pajisjet elektronike. Lejohen, vetëm për diskutime.



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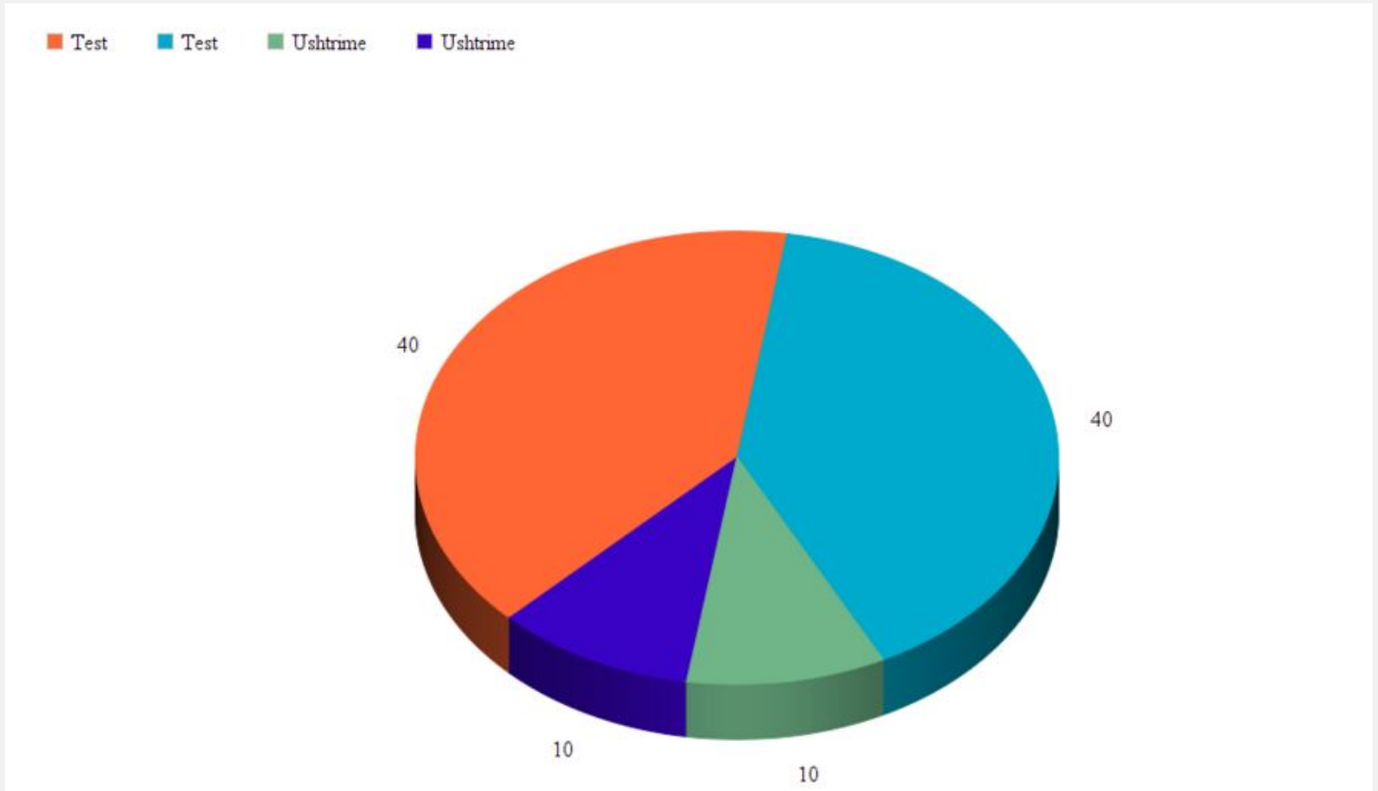
Notimi dhe detyrat

Ushtrimet. 20%

- Javën e tretë dhe të nëntë.
- 2x nga 10%.

Testet. 80%

- Dy herë përgjatë semstrit
- Secili test me vlerë nga 40%.



Provimet. 100%

- Mbahet gjatë afatit të rregulltë
- Vlera e pikëve nëpër teste nuk vlenë.

Literatura (textbook)

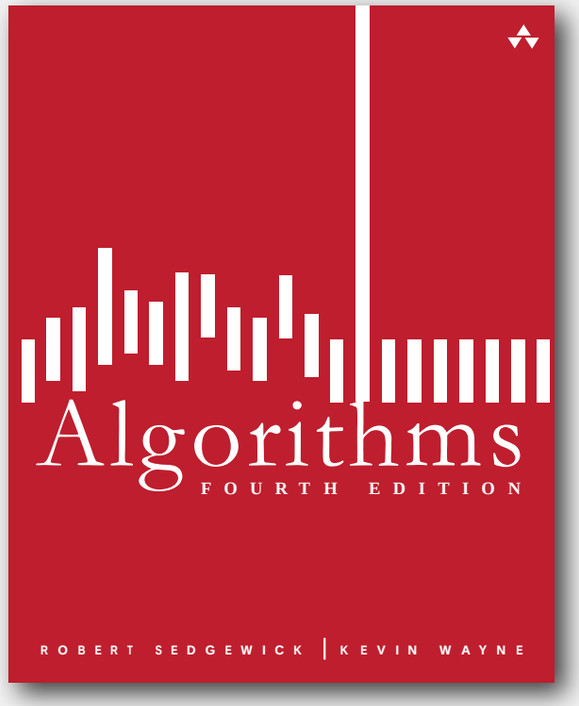
Leximi. Algorithms 4th edition by R. Sedgewick and K. Wayne, Addison-Wesley Professional, 2011, ISBN 0-321-57351-X.

1st edition (1982)

2nd edition (1988)

3rd edition (1997)

3rd book scanned by Google books

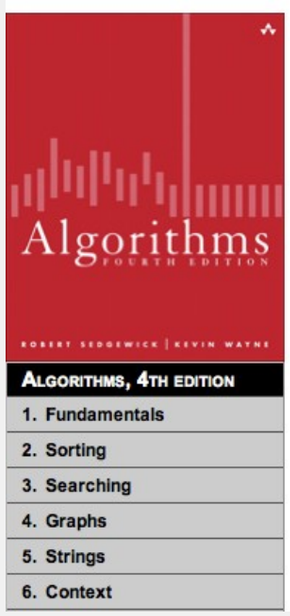


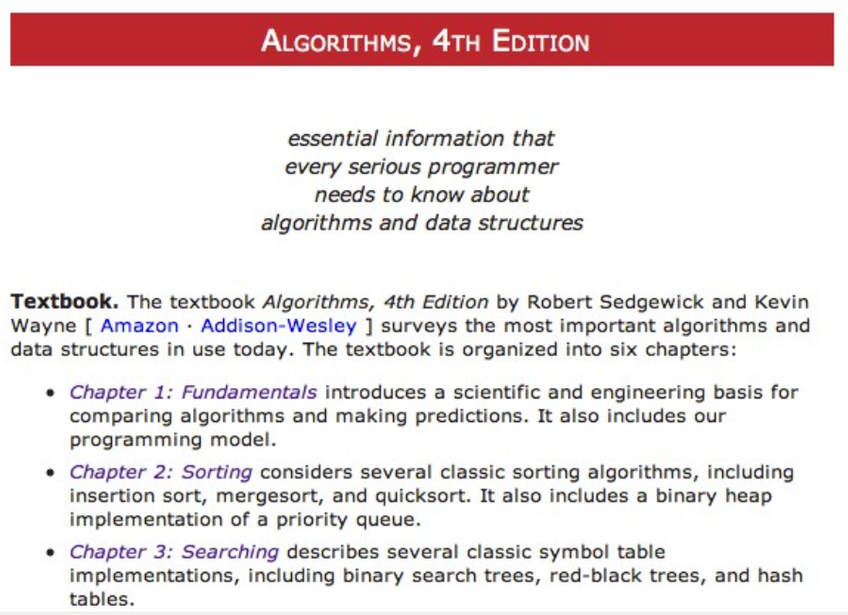
4th edition (2011)

Ligteratura (textbook)

Booksite.

- Përmbledhjet e ligjëratave.
- Download code prej librit.
- APIs dhe Javadoc.





http://algs4.cs.princeton.edu